

I Year I Semester

PHYSICS FOR MECHANICAL SYSTEMS (Mechanical Engg.)**(AI Integrated)**

Course Code	Year/Sem	L	T	Pr	P	Credits
26APH03T	I Year I Sem	3	0	0	0	3

Prerequisites: Engineering Mechanics, Engineering Mathematics, Basic Physics**Course Objectives**

The objectives of this course:

1. To provide students with a strong foundation in wave optics concepts such as interference, diffraction and polarization along with their applications in optical measurements, dimensional inspection and basic AI-assisted image and fringe pattern analysis techniques.
2. To introduce the fundamentals of crystallography and X-ray diffraction methods for understanding crystal structures, phase identification and material characterization using simple AI-assisted data analysis approaches.
3. To equip students the basic principles of quantum mechanics and quantum computing, including quantum states, tunneling, superposition and secure communication with introductory AI-assisted quantum simulations and visualization tools.
4. To familiarize students with oscillations, vibration control systems, lasers and precision engineering applications along with basic AI-assisted monitoring, predictive maintenance and intelligent control concepts used in modern industries.
5. To impart knowledge on the principles of ultrasonics and non-destructive testing methods for defect detection, quality inspection and structural analysis using simple AI-assisted signal interpretation and automated inspection techniques.

Course Content**UNIT I: Applied Wave Optics****9 hrs****Core Content:**

Interference: Principle of superposition and coherence, Interference, Types of interference, Interference in thin films (reflection geometry) and Newton's rings. Applications: Determination of wavelength and refractive index.

Diffraction and Optical Surveying: Introduction to diffraction; Fresnel and Fraunhofer diffraction. Fraunhofer diffraction due to single, double and N-slits (Diffraction Grating), Dispersive power and Resolving power of Grating. Application: 3D Scanning for Dimensional Inspection of Casted Parts.

Polarization: Types of polarization, Nicol's prism, Half-wave and Quarter-wave plates.

AI Integration:

AI-integrated holographic interferometry for micro-deformation analysis. Python-based fringe pattern analysis using image processing (OpenCV).

Free/Open-Source AI Tools:

Python (NumPy, SciPy, Matplotlib and scikit-learn).

UNIT II: Crystallography and AI-Driven Material Engineering**8 hrs****Core Content:**

Crystallography: Space lattice, Basis, Unit Cell and lattice parameters – Crystal systems, Bravais Lattices — Coordination number - Packing fraction of SC, BCC & FCC - Miller indices – Separation between successive (h k l) planes.

X-ray diffraction: Bragg's law - X-ray Diffractometer – Crystal structure determination by Laue's method. Application: XRD Pattern Analysis for Phase Identification, Determination of Crystal Structure of Metals and Alloys

AI Integration:

AI-assisted XRD pattern analysis for phase identification and crystal structure determination. ML-based classification of diffraction patterns using scikit-learn.

Free/Open-Source AI Tools:

Python (NumPy, SciPy, Matplotlib and scikit-learn).

UNIT III: Introduction to Quantum Mechanics & Quantum Computing**10 hrs****Core Content:**

Introduction to quantum mechanics, de-Broglie concept (matter waves), Heisenberg's Uncertainty Principle, Wave function, Postulates of Quantum Mechanics, Schrödinger's time-independent wave equation, Particle in a one-dimensional potential well, Quantum tunnelling (basic idea).

Introduction to Quantum Computing: Classical bits vs qubits. Concept of superposition and Entanglement, Basic Quantum Operations: Hadamard gate (idea of superposition), Pauli-X gate. Basic idea of quantum measurement, Applications: Basic idea of quantum cryptography and secure communication.

AI Integration: AI-assisted quantum circuit simulation using IBM Qiskit (free). Python-based visualization of wave functions and quantum probability densities

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT IV: Oscillations, Vibration Control and Laser-Based Metrology**10 hrs****Core Content:**

Introduction to oscillations and oscillations of springs. Series and parallel combinations of springs in Suspension Systems, Stiffness factor and its physical significance, vibration isolation. Theory of damped oscillations (Over, Under and Critical damping – all qualitative). Forced oscillations (qualitative) and Resonance; Sharpness of resonance and Quality factor (Q). Applications (Qualitative): shock absorbers, noise, and vibration reduction.

Lasers and Precision Engineering: Principles of Absorption, Spontaneous and Stimulated Emission, Population Inversion. Three-level and Four-level laser systems. Construction and working of He-Ne Laser and Nd-YAG Laser. Applications (qualitative only): Role of high-power lasers in CNC Laser Cutting and Welding.

AI Integration: AI-Assisted Adaptive Suspension System for EVs, AI-Based Vibration Monitoring and Predictive Maintenance.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT V: Ultrasonics and Non-Destructive Testing (NDT)**8 hrs****Core Content:**

Introduction to ultrasonics; Production of ultrasonics: Piezoelectric method and Magnetostriction method, Detection of ultrasonics: Kundt's tube and Quartz crystal methods. Applications: Use of Ultrasonic Flaw Detectors to identify internal cracks, voids and thickness variations in cast and forged engine components.

Non-Destructive Testing (NDT): Introduction, need and advantages in mechanical engineering structures, Basic inspection methods: Visual inspection and AI-based drone inspections for inaccessible materials (qualitative). Liquid penetrant testing: basic principle for detecting surface defects in steel. Applications: Detecting surface and sub-surface defects, sorting alloys and measuring coating thickness on automotive parts.

AI Integration: AI-recognition in abnormal echo patterns, AI-assisted additional reflections and frequency distortions.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

Course Outcomes (COs)

After completion of the course, student will be able to

CO	Course Outcomes	Bloom's Level
CO1	Examine the principles of interference, diffraction and polarization and utilise basic AI-assisted optical analysis techniques in engineering applications.	L3 (Apply)
CO2	Analyze crystal structures, diffraction patterns and material properties using X-ray diffraction methods and basic AI-assisted material characterization tools.	L4 (Analyze)
CO3	Interpret the concepts of quantum mechanics and quantum computing and apply basic AI-assisted quantum simulation techniques for secure communication systems.	L3 (Apply)
CO4	Analyze oscillatory systems, vibration control mechanisms and laser-based metrology techniques using basic AI-assisted monitoring and predictive approaches.	L4 (Analyze)
CO5	Evaluate ultrasonic testing and non-destructive testing methods for defect identification and quality assessment using basic AI-assisted signal analysis techniques	L5 (Evaluate)

CO–PO Articulation Matrix

Scale: 3 – Strong, 2 – Moderate, 1 – Slight, – No Correlation

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	1	2	3	-	-	-	-	-	-
CO2	3	3	2	2	3	-	-	-	-	-	-
CO3	3	2	2	2	3	-	-	-	-	-	-
CO4	3	3	2	2	3	-	-	-	-	-	-
CO5	3	3	2	3	3	-	-	-	-	-	-

Textbooks

1. Avadhanulu, Kshirsagar & Arun Murthy, Engineering Physics, S. Chand, 2019.
2. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford Press, 2015.

Reference Books

3. B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2021.
4. Shatendra Sharma & Jyotsna Sharma, Engineering Physics, Pearson, 2018.
5. Sanjay D. Jain, D. Sahasrabudhe & Girish, Engineering Physics, Universities Press, 2010.
6. M. R. Srinivasan, Engineering Physics, New Age International, 2009.
7. S. O. Pillai, Solid State Physics, New Age International.
8. R. Murugesan & Kiruthiga Sivaprasath, Modern Physics, S. Chand.

Web Resources & NPTEL Links

- NPTEL – nptel.ac.in
- MIT OpenCourseWare – ocw.mit.edu
- HyperPhysics – hyperphysics.phy-astr.gsu.edu
- Feynman Lectures – feynmanlectures.caltech.edu
- PhET Simulations – phet.colorado.edu

Recommended Free & Open-Source AI/Computational Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) – FREE on Google Colab for wave optics and signal analysis
- TensorFlow / Keras (free, open-source) – For crack and defect detection CNNs on Google Colab
- scikit-learn (free, open-source) – For anomaly detection in vibration and NDT data
- IBM Qiskit (free, open-source) – For quantum computing simulations on Google Colab
- OpenCV (free, open-source) – For image processing in visual inspection and interferometry
- PhET Interactive Simulations (free) – For wave optics, quantum mechanics and oscillations

Course Code	Title	L	T	P	C
26APH03P	PHYSICS FOR MECHANICAL SYSTEMS LAB (AI INTEGRATED)	0	0	3	1.5

Course Objectives

1. To determine optical properties—including dispersive power, thin-film/object thickness, lens radius of curvature, and mercury spectral wavelengths—using prism spectroscopy, interference setups, and diffraction gratings (Exp 1–4).
2. To analyze laser diffraction dynamics by measuring laser emission wavelengths and estimating microscopic particle sizes (Exp 5–6).
3. To evaluate quantum photon interaction dynamics and photoelectric emission parameters by estimating Planck's constant (Exp 7).
4. To characterize elastic deformation, vibrational coupling, mass-spring oscillations, and dynamic rigidity using Hooke's law, spring combinations, coupled oscillators, and torsional pendulums (Exp 8, 11–13, 15).
5. To evaluate acoustic wave dynamics, resonance, and sound velocity across solid wires and liquid media using Melde's setup, sonometer laws, and acoustic gratings (Exp 9, 10, 14).

List of Experiments (Any ten experiments need to be performed)

Part A – Regular Experiments

1. Determination of the dispersive power of the prism.
2. Determination of the thickness of a thin object by the wedge method
3. Determination of the radius of curvature of the lens by Newton's rings
4. Determination of wavelengths of various colours of mercury spectrum using a diffraction grating in the normal incidence method
5. Determination of wavelength using a diffraction grating
6. Determination of particle size using a diode laser
7. Estimation of Planck's constant using the photoelectric effect
8. Determination of spring constant by Hooke's law
9. Melde's experiment: Determination of the frequency of the tuning fork
10. Sonometer: Verification of the three laws of stretched strings
11. Determination of the frequency of normal modes of the Coupled Oscillator
12. Determination of spring constant – series and parallel combinations.
13. Study of oscillations of a mass under different combinations of springs (Loaded Springs)
14. Determination of ultrasonic velocity in liquid (Acoustic grating)
15. Determination of Rigidity modulus of the material of a wire, dynamic method (Torsional pendulum)

Part B – AI Integration Lab Exercises

Any **three** of the following AI exercises must be completed using Python / Google Colab or appropriate AI/LLM tools.

1. **Oscillation & Vibration Signal Automation (Python / Matplotlib):** input time-vs-displacement data from the sonometer, Melde's, or coupled-oscillator experiment; fit a sinusoidal / damped-sinusoidal model to the signal; extract frequency, amplitude, and damping coefficient; overlay the fitted curve on the experimental data points.
2. **Spring-Constant & Elastic Property Analysis (Python / NumPy):** input load vs. extension data from the Hooke's-law or loaded-springs experiment; apply linear regression to extract the spring constant; automate computation of percentage error and R^2 against the expected value.
3. **Vibration Anomaly / Defect Detection (Python / scikit-learn or TensorFlow):** use vibration or acoustic-grating signal data (or a sample vibration/NDT dataset) to train a simple anomaly-detection model or CNN-based crack/defect classifier; evaluate classification accuracy and discuss relevance to condition monitoring of mechanical components.

AI Integration

AI integration adds two to three exercises after the regular experiments: (i) Python automation of oscillation/vibration signal analysis, fitting sinusoidal or damped-sinusoidal models to extract frequency and damping using Matplotlib/NumPy; (ii) regression-based extraction of the spring constant and elastic parameters from Hooke's-law/loaded-spring data; (iii) an introductory scikit-learn/TensorFlow exercise applying anomaly-detection or CNN-based classification to vibration or acoustic-grating signals — reinforcing the connection between the physical experiment and data-driven condition monitoring, and giving Mechanical Engineering students hands-on practice applying programming and machine-learning skills to experimental mechanics and NDT.

Free / Open-Source AI Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) on Google Colab (free) — wave-optics automation and vibration/signal analysis
- TensorFlow / Keras (free, open-source) — crack- and defect-detection CNNs on Google Colab
- scikit-learn (free, open-source) — anomaly detection in vibration and NDT data
- PhET Interactive Simulations (free) — wave optics, quantum mechanics, and oscillations: phet.colorado.edu
- Virtual Labs — IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences — online simulation of optics and mechanical-vibration experiments
- HyperPhysics (free) — concept reference for optics, quantum theory, and oscillations: hyperphysics.phy-astr.gsu.edu

Web Resources

- HyperPhysics — hyperphysics.phy-astr.gsu.edu
- PhET Simulations — phet.colorado.edu
- NPTEL — nptel.ac.in
- Virtual Labs (IIT Bombay) — iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences

Course Outcomes (COs)

After successful completion of this lab course, the student will be able to:

CO	Course Outcomes	Bloom's Level
CO1	Compute optical parameters—including dispersive power, object thickness, radius of curvature, and spectral wavelengths—using interference, diffraction, and refractometry techniques.	L3 (Apply)
CO2	Analyze laser diffraction patterns to determine monochromatic light wavelengths and evaluate microscopic particle dimensions.	L3–L4 (Apply, Analyse)
CO3	Analyze quantum photoelectric emission behavior to estimate Planck's constant from stopping potential measurements.	L3–L4 (Apply, Analyse)
CO4	Evaluate spring constants, normal mode frequencies, dynamic oscillatory behavior, and material rigidity moduli across various mechanical systems.	L4–L5 (Analyse, Evaluate)
CO5	Evaluate sound wave velocity, standing wave resonance, and ultrasonic wave propagation in solids and liquids against theoretical models.	L5–L6 (Evaluate, Create)

CO – PO Articulation Matrix (NBA)

Scale: 3 – Strong | 2 – Moderate | 1 – Slight | – No Correlation

CO	Bloom's Lvl	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11
CO1	L3	3	2	2	3	2	–	–	–	–	–	–
CO2	L3–L4	3	3	2	3	3	–	–	–	–	–	–
CO3	L4	3	3	2	3	2	–	–	–	–	–	–
CO4	L3–L4	3	2	2	3	2	–	–	–	–	–	–
CO5	L5–L6	2	3	3	3	3	–	–	–	–	–	–

Recommended Free & Open-Source AI / Computational Tools

Python (NumPy, SciPy, OpenCV, Matplotlib) on Google Colab (free) – optics automation, fringe analysis, photoelectric regression, EM simulations

scikit-learn (free, open-source) – semiconductor data analysis and signal classification

OpenEMS (free, open-source) – electromagnetic field simulation and antenna design

PhET Interactive Simulations (free) – wave optics, semiconductors, and quantum phenomena: phet.colorado.edu

Virtual Labs – IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences – online simulation of optics, semiconductor and EM experiments

HyperPhysics (free) – concept reference for optics, quantum and EM theory: hyperphysics.phy-astr.gsu.edu

LibreOffice Calc (free, open-source) – standardised data recording templates for all experiments
--

Note: Lab record to be submitted for every experiment at the next session. AI exercises to be submitted as Google Colab (.ipynb) notebooks.

Reference Books

1. M. N. Avadhanulu & P. G. Kshirsagar, A Textbook of Engineering Physics, S. Chand & Company, 2019.
2. H. K. Malik & A. K. Singh, Engineering Physics, Tata McGraw-Hill, 2010.
3. R. K. Gaur & S. L. Gupta, Engineering Physics, Dhanpat Rai Publications, 8th Edition, 2012.
4. B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2012.
5. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford University Press, 2015.